Algebra and Discrete Mathematics (ADM)

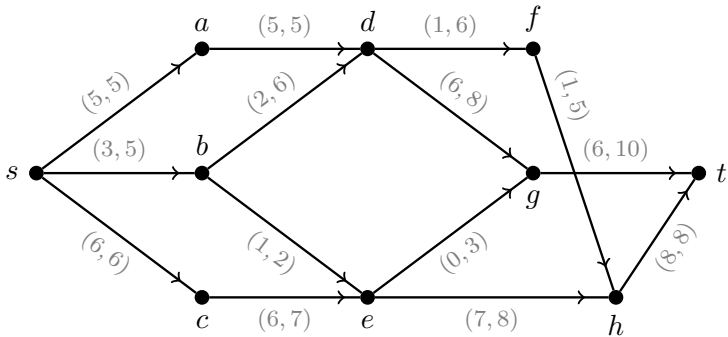
Tutorial 11 Trees and networks

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Augmenting Flow Algorithm – steps

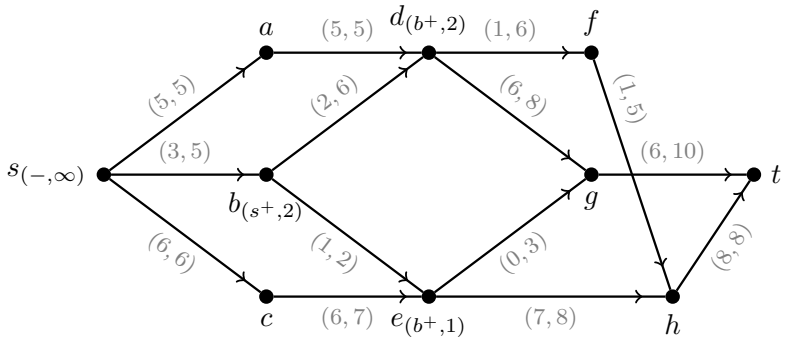
1. Label s with $(-, \infty)$, set $\sigma(v) = \infty$ for other vertices
2. Choose a labeled vertex x, scan x:
   a. For any arc yx, if $f(yx) > 0$ and y is unlabeled, then label y with $(x^-, \sigma(y))$, where $\sigma(y) = \min\{\sigma(x), f(yx)\}$
   b. For any arc xy, if $k(xy) > 0$ and y is unlabeled, then label y with $(x^+, \sigma(y))$, where $\sigma(y) = \min\{\sigma(x), k(xy)\}$
3. If t has been labeled, go to Step 4. Otherwise, choose another labeled vertex that has not yet been scanned and return to Step 2. If all labeled vertices have been scanned and t is still unlabeled, then f is a maximum flow.
4. Find an s-t chain K by backtracking from t to s using the vertex labels. Along the edges of K, increase the flow by $\sigma(t)$ units if they are in the forward direction and decrease by $\sigma(t)$ units in they are in the backward direction. Remove all vertex labels except that of s and return to Step 2

Augmenting Flow Algorithm



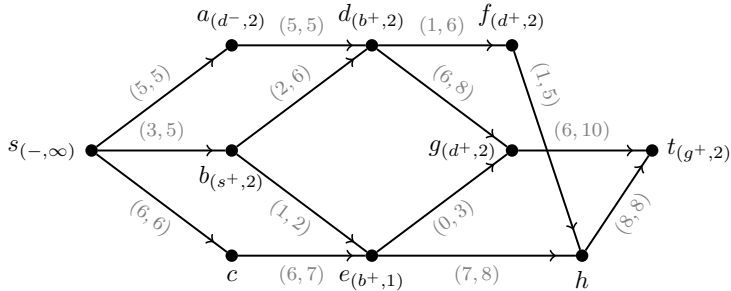
- Example: Question 3(1)

Augmenting Flow Algorithm



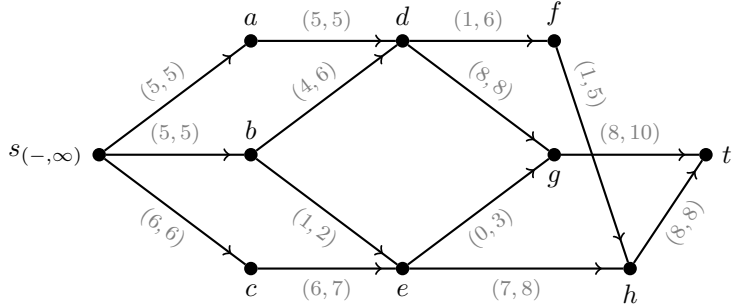
- Step 1. Label s
- Step 2. Label b
- Step 3. Choose b
- Step 2. Label d and e

Augmenting Flow Algorithm



- Step 3. Choose d
- Step 2. Label a, f, and g
- Step 3. Choose g
- Step 2. Label t

Augmenting Flow Algorithm



- Step 3. Since t is labeled, go to Step 4
- Step 4. Backtracking gives the chain $sbdgt$
  - Increase the flow by $\sigma(t) = 2$ units along each of these edges since all are in the forward direction.
  - Update the network flow and remove all labels except for s
- No further vertices can be labeled
- Value of flow $|f| = f^+(s) = 16$

Min-Cut Method

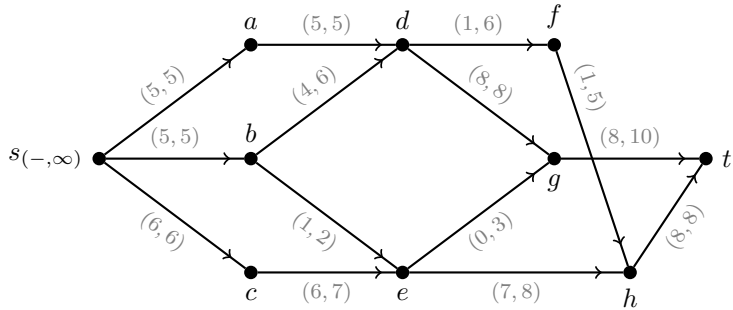
Steps

1. Let $G = (V, A, c)$ be a network with a designated source s and sink t and each arc is given a capacity c
2. Apply the Augmenting Flow Algorithm
3. Define an $s - t$ cut $(P, \overline{P})$ where P is the set of labeled vertices from the final implementation of the algorithm
4. $(P, \overline{P})$ is a minimum $s - t$ cut for G

Note

In practice, we can perform the Augmenting Flow Algorithm and the Min-Cut Method simultaneously, thus finding a maximum flow and providing a proof that it is maximum (through the use of a minimum cut) in one complete procedure.

Augmenting Flow Algorithm



- Value of flow $|f| = f^+(s) = 16$
- $P = \{s\}$
- $c(P, \overline{P}) = 5 + 5 + 6 = 16$